

GeoReach

Geospatial Accessibility Intelligence for Flood Response

*Which villages get cut off when Assam floods — **and who to reach first.***

WHY IT MATTERS

Every monsoon, the Brahmaputra severs road links between remote settlements and the hospitals that serve them. Relief teams lack a fast, evidence-based map of which places are isolated — and where to send boats first.

FOCUS AREA

Dhemaji

Lakhimpur

Majuli — *world's largest river island*

75+

lives lost

700K

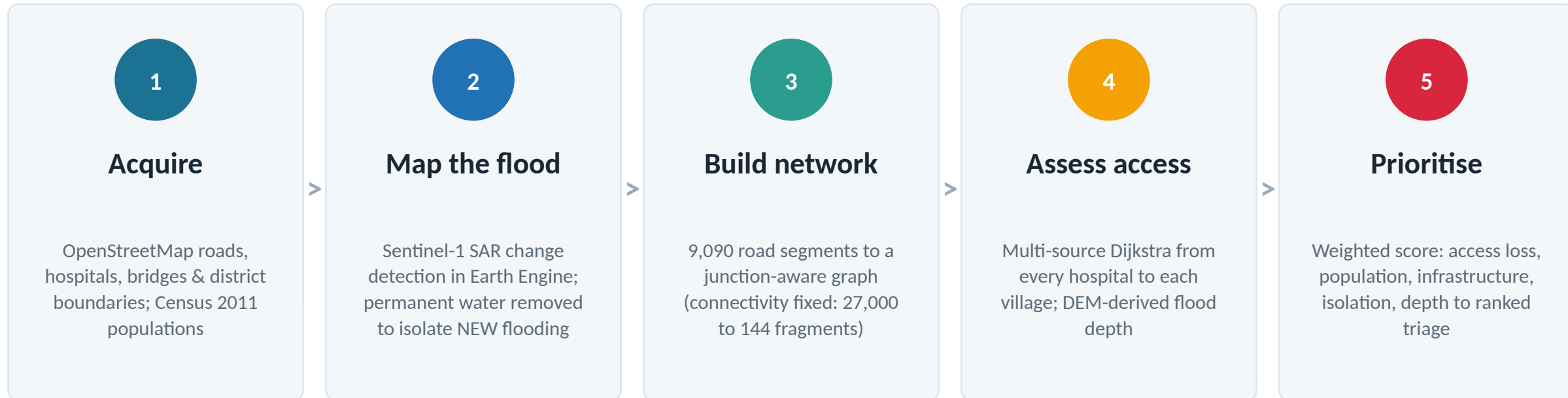
people displaced

900+

villages submerged

Engineering Workflow

A reproducible pipeline from raw open data to a decision-ready map — end to end in ~23 seconds.



BUILT ENTIRELY ON OPEN / PUBLIC DATA

Sentinel-1 SAR

SRTM DEM

OpenStreetMap

Census 2011

CWC gauges

ASDMA reports

Python • GeoPandas • NetworkX • rasterio • Folium/Leaflet

Results — Observed 2026 Flood

Driven by real Sentinel-1 satellite flood extent over the three districts.

22%

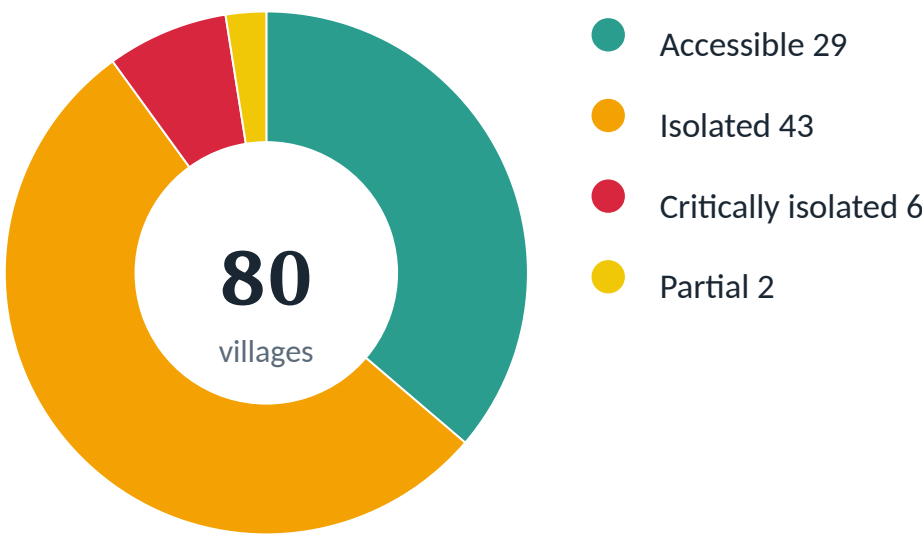
of roads cut off

2,013 of 9,090 segments

49

settlements isolated

6 critically isolated



807 km²

area flooded

TOP PRIORITY TO REACH FIRST

- | | | | | | | | | | |
|---|--|---|---|---|--|---|---|---|---|
| 1 | Jonai
pop 15,420 · critically isolated | 2 | Gogamukh
pop 12,750 · critically isolated | 3 | Dhakuakhana
pop 12,800 · critically isolated | 4 | Kamalabari
pop 12,200 · critically isolated | 5 | Bihpuria
pop 10,500 · critically isolated |
|---|--|---|---|---|--|---|---|---|---|

Innovation & Impact

S

Real satellite flood

Sentinel-1 SAR sees through monsoon cloud; permanent-water exclusion isolates genuinely new inundation

E

Terrain-derived depth

Flood depth computed from SRTM elevation inside each flooded area — not a flat assumption

N

Junction-aware routing

Rebuilt the road graph so intersections connect (27,000 to 144 fragments), making isolation real

T

Multi-criteria triage

Transparent weighted scoring turns raw geography into a ranked, defensible action list

IMPACT

Turns days of manual assessment into a 23-second, repeatable run. Gives disaster authorities an interactive map of who is cut off and where to send boats first — transferable to any flood-prone district, built entirely on free, public data.